



Web Development Foundations

Week 3 - "Layout: Control Space"
Day 1: "Layout Foundations"

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Lab Review: Successful Path

One successful path:

- ▶ one shared stylesheet
- ▶ readable text
- ▶ consistent navigation
- ▶ intentional color and spacing
- ▶ class-based styling

Today we keep the CSS layer and improve the space.



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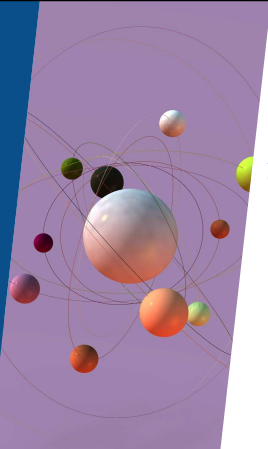


Layout Is Structure In Space

A web page is not just content plus color.

Layout answers:

- ▶ what belongs together?
- ▶ what should be easy to scan?
- ▶ where does the reader's eye go first?
- ▶ what happens when the screen changes?

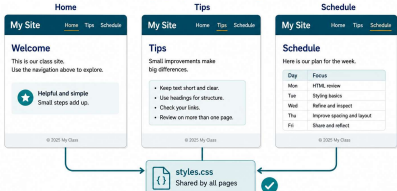


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Week 2: One Successful Styled Website Path

Three pages, one shared stylesheet.



One successful path. Not the only answer. It's okay to revise, try again, and keep improving.

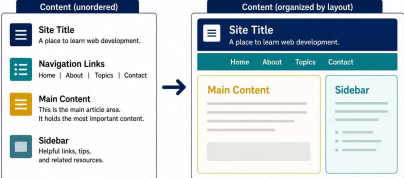
Assignment 2 Success Path

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Layout: Structure Expressed in Space

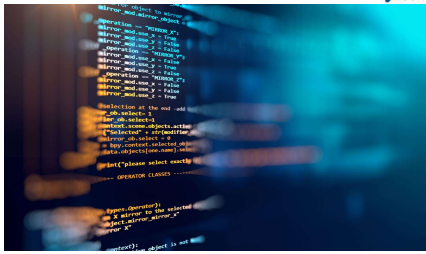
The same content. Organized into readable zones.



Layout makes relationships visible.

Layout Is Structure In Space

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Review Coding Lab

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From Styled Page To Organized Page

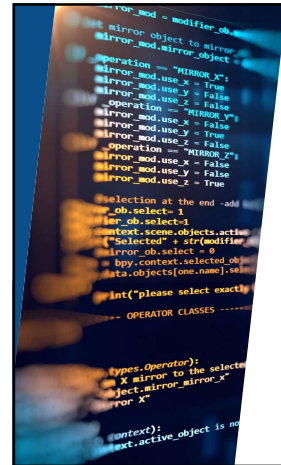
Styling changes appearance. Layout organizes space.

A styled page may still feel:

- ▶ crowded
- ▶ scattered
- ▶ hard to scan
- ▶ awkward on different screens

Layout makes relationships visible.

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What We Will Use Today

- ▶ box model
- ▶ layout vocabulary
- ▶ 12-column grid idea
- ▶ content, padding, border, margin
- ▶ container
- ▶ `display: flex`
- ▶ `gap`
- ▶ `flex`
- ▶ save, refresh, resize

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Styled vs. Organized Layout

Layout is about structure and space, not just decoration.

Styled

My Site

Welcome

About Us

Latest Tips

Contact

Organized

My Site

Welcome

About Us

Latest Tips

Contact

Same content. Better spatial relationships.

From Styled Page To Organized Page

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Today's Beginner Layout Toolbox

The essential tools for building simple, effective layouts.



Today's Toolbox

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What Counts as Success Today

- you can point to the boxes on a page
- you can explain margin, padding, and border
- one Flexbox container changes layout
- the page becomes easier to scan
- you notice what still needs responsive work

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What We Will "Skip" For Now

- ▶ full responsive design strategy
- ▶ many Flexbox properties
- ▶ implementing 960.gs or grid frameworks
- ▶ modern CSS Grid module
- ▶ frameworks like Bootstrap
- ▶ JavaScript behavior

Today: recognize grid thinking; implement one Flexbox layout.

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Week 3: Layout Topics Parked for Later

We're building one step at a time.

Parked for Later

- full responsive strategy: Plan for all screen sizes with a complete approach.
- many Flexbox properties: Explore and practice additional Flexbox controls.
- implementing 960.gs: Use the 960 Grid System to speed up development.
- modern CSS Grid module: Learn the full power of CSS Grid for layouts.
- frameworks: Use pre-built systems to speed up development.
- JavaScript layout: Add dynamic layout control with JavaScript.

Recognize grid thinking. Implement one Flexbox layout.

Parked For Later

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Margin And Padding Are Not The Same

Padding adds space inside the element.
Margin adds space outside the element.

Quick check:

- ▶ text too close to its own edge? padding
- ▶ two boxes too close together? margin or gap
- ▶ layout feels cramped? inspect the space before adding more color

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The Box Model Is The First Layout Map

Every element takes up space.

Think of an element as:

- ▶ content: the actual text or image
- ▶ padding: breathing room inside the box
- ▶ border: the edge of the box
- ▶ margin: space outside the box

Layout begins when you can see the boxes.

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Margin vs. Padding

Two kinds of space. Two different jobs.

Padding = space inside
Padding adds space between the content and the edge of the box.

Margin = space outside
Margin adds space outside the box, separating it from other elements.

Inside space and outside space solve different problems.

Margin Versus Padding

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The CSS Box Model

Every element is a box.

margin: Space outside the border.

border: The edge around the padding and content.

padding: Space between the content and the border.

content: The text, images, or elements.

Layout begins when you can see the boxes.

Box Model Map

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Layout Systems In The Reading

Duckett's layout chapter introduces several layout ideas:

- ▶ positioning moves elements from normal flow
- ▶ fixed layouts use set widths
- ▶ liquid layouts stretch with the browser
- ▶ grid systems align content into columns

Read these as vocabulary and history.
Do not try to implement all of them this week.

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Four Layout Ideas from Our Reading

Different ways to organize content on a web page.

1
positioning

Place elements exactly where you want them on the page.

2
fixed layout

Set a specific width. The layout stays the same size.

3
liquid layout

Use percentages. The layout stretches or shrinks with the window.

4
grid system

Use columns to line up content and create consistent layouts.

Read these as layout vocabulary and history.

Layout Systems

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Containers Control Groups

Flexbox starts with a container.
The container decides how its children line up.

Example:

- ▶ ``.page`` is the container
- ▶ ``.content`` and ``.sidebar`` are children
- ▶ the layout rule goes on ``.page``

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Grid Thinking, Not Grid Implementation

The 12-column grid idea helps designers plan alignment.

Useful idea:

- ▶ line up content consistently
- ▶ divide space into columns
- ▶ make pages feel organized

This week:

- ▶ recognize the grid-planning idea
- ▶ do not build a 960.gs layout
- ▶ use Flexbox for the lab implementation

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Flexbox: Parent-Child Idea

The container controls how its children are arranged.

Put the layout rule on the container.

Container Controls Children

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12-Column Grid Thinking

Use a simple grid idea to plan alignment, then build with Flexbox.

12-column grid idea = plan alignment

The grid is a planning guide to keep things aligned and balanced.

960.gs = recognize, do not build today

A classic 12-column CSS framework. Study this idea, not the implementation. Recognize the concept. Don't build the framework.

Flexbox = lab implementation

Use Flexbox to build your layout.

Display: flex; Gap between items

Grid thinking helps planning. Flexbox is today's build tool.

Grid Thinking

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Flexbox, Gently

Flexbox can organize children in a row or column.

Today's small pattern:

```
.page {
  display: flex;
  gap: 16px;
}
```

Start with one container. Then inspect what changed.

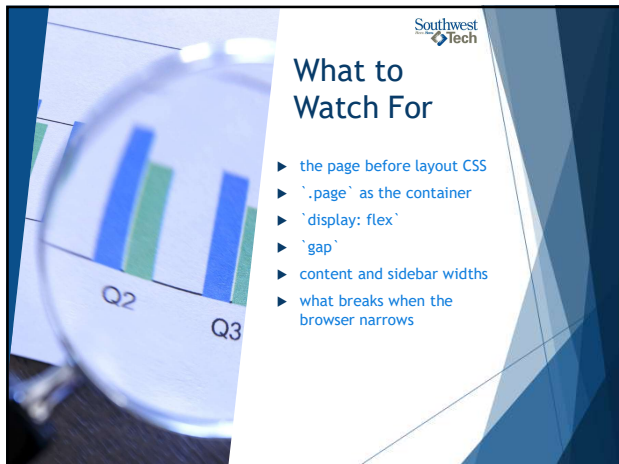
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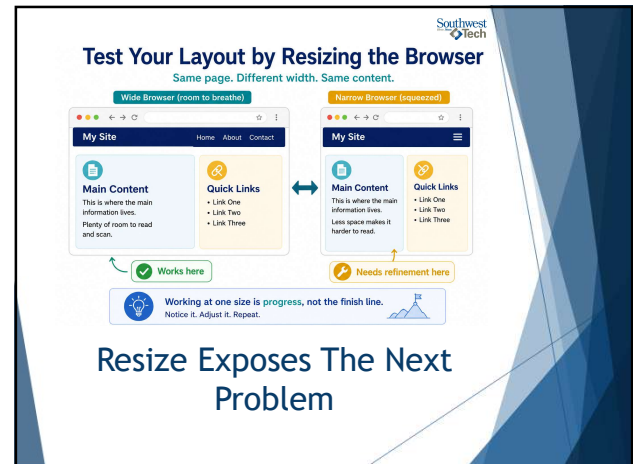
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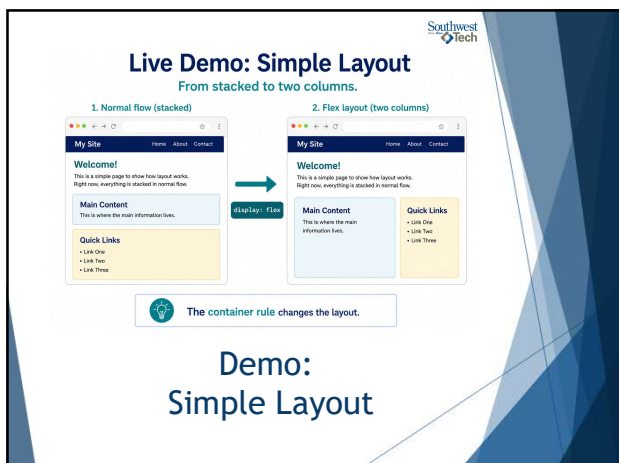
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Evidence

- ▶ HTML and CSS files still render
- ▶ Flexbox appears in your CSS
- ▶ before/after screenshot if possible
- ▶ one sentence explaining the layout change
- ▶ note one problem you still need to improve

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Closing Success Check

Today:

- ▶ boxes became visible
- ▶ spacing became intentional
- ▶ one container controlled a group
- ▶ Flexbox created a first layout

Next:

- ▶ *We make layout respond when the screen changes.*

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Thank you!

Have Fun!

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